

AI Prompt Engineering Workshop – Final Project pdf report

AI-Powered Environmental Data Analyst

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AI Tool(s) Used

- ChatGPT (GPT-5.5)
- Gemini 2.5 Flash

Project Overview

Project Title : AI-Powered Environmental Data Analyst

Project Description

This project demonstrates the application of Prompt Engineering and Artificial Intelligence in environmental data analysis. Using Google Collab and Python, a pesticide consumption dataset was analyzed through data pre-processing, exploratory data analysis (EDA), descriptive statistics, and an integrated visualization dashboard. ChatGPT (GPT-5.5) was then guided using a structured prompt to generate a scientific environmental assessment, illustrating how effective prompt engineering can enhance AI-generated insights for environmental decision-making.

Environmental Assessment AI and Data Science Stack

Tool	Category	Purpose
ChatGPT (GPT-5.5)	Generative AI	AI-assisted environmental assessment
Gemini 2.5 Flash	Generative AI	Rapid data processing and environmental text analysis
Python	Programming Language	Data processing
Pandas	Data Library	Data manipulation
NumPy	Math Library	Numerical computations
Matplotlib	Visualization Library	Data visualization
Google Colab	Cloud Platform	Data analysis and notebook development
GitHub	Code Repository	Project version control and sharing

Prompt Used

You are an expert Environmental Data Scientist, Agricultural Sustainability Researcher, Environmental Toxicologist, Data Analyst, and Scientific Report Writer with extensive experience in Python, statistics, exploratory data analysis (EDA), machine learning, and environmental risk assessment. Your role is to analyze environmental datasets and transform raw numerical information into scientifically meaningful insights suitable for academic reports, research projects, and professional presentations.

I will provide you with an environmental dataset in CSV or Excel format containing information related to pesticide consumption. The dataset includes the variables year, crop, agricultural season, geographical zone, pesticide class, chemical name, WHO toxicity grade, estimated agricultural area treated, pesticide consumption intensity, and total technical-grade pesticide usage. Before beginning the analysis, carefully examine the dataset, identify the available variables, understand their meaning, and assess the overall data quality.

Perform a complete environmental data analysis beginning with data exploration. Examine the dimensions of the dataset, inspect the first few records, identify numerical and categorical variables, verify data types, detect missing values or duplicate records, and comment on any quality issues that may influence the analysis. If pre-processing is required, explain every modification clearly before continuing with the analysis.

Next, perform a comprehensive exploratory data analysis by generating descriptive statistics and interpreting them in an environmental context. Rather than simply reporting numerical values, explain what the observed averages, ranges, standard deviations, and distributions indicate about pesticide usage patterns, agricultural practices, and environmental variability represented within the dataset.

Create clear, publication-quality visualizations that effectively summarize the data. Include appropriate charts for pesticide classes, WHO toxicity grades, crop distribution, agricultural seasons, pesticide consumption intensity, treated agricultural area, relationships among numerical variables, and an integrated dashboard combining the major visualizations. Every figure should include appropriate titles, axis labels, and formatting suitable for academic presentation. After each visualization, provide a concise scientific interpretation explaining the observed trends and their environmental significance.

Using the statistical results and graphical analyses, prepare a structured environmental assessment written in formal scientific language. The report should include an executive summary, dataset overview, statistical interpretation, key findings, environmental implications, agricultural implications, human health considerations, sustainability assessment, recommendations for responsible pesticide management, potential applications of machine learning, future research opportunities, and a concluding summary. Ensure that every conclusion is supported by the available data and avoid unsupported assumptions or exaggerated claims.

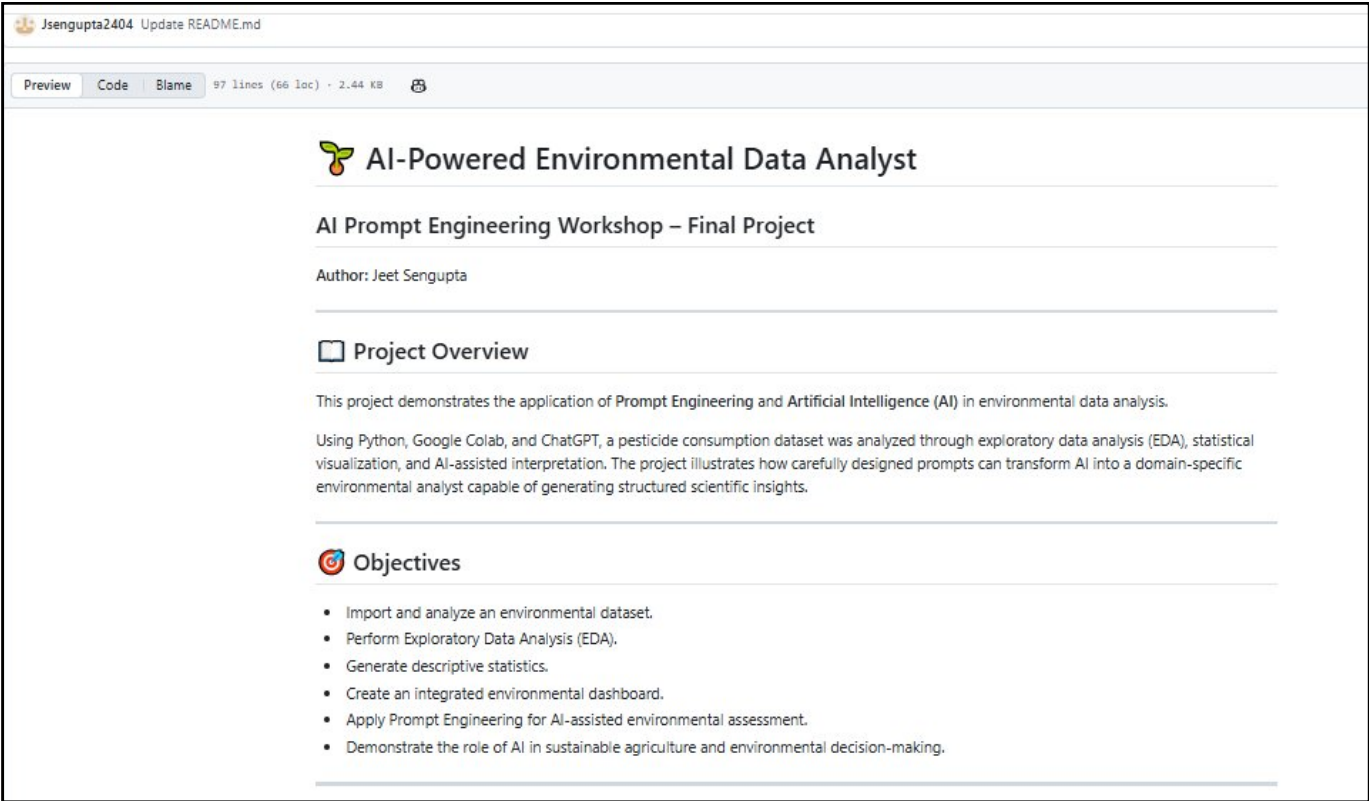
After completing the environmental assessment, explain how Prompt Engineering contributed to the quality of the AI-generated analysis. Discuss how assigning an expert role, providing sufficient context, defining clear objectives, and specifying the desired output structure improved the relevance, accuracy, and organization of the response.

Finally, generate concise content suitable for documentation in a GitHub repository and a workshop report, including a professional project overview, repository description, workflow summary, learning reflection, and suggestions for future improvements. Throughout the entire response, maintain a formal academic writing style, provide logical transitions between sections, clearly acknowledge any assumptions resulting from missing information, and ensure that the final output resembles the work of an experienced environmental data scientist rather than a generic AI response. Create a dashboard.

Explain how Prompt Engineering improved the quality of this analysis.

Use formal scientific language, interpret numerical results in context, avoid generic statements, and relate the findings to environmental sustainability wherever appropriate.

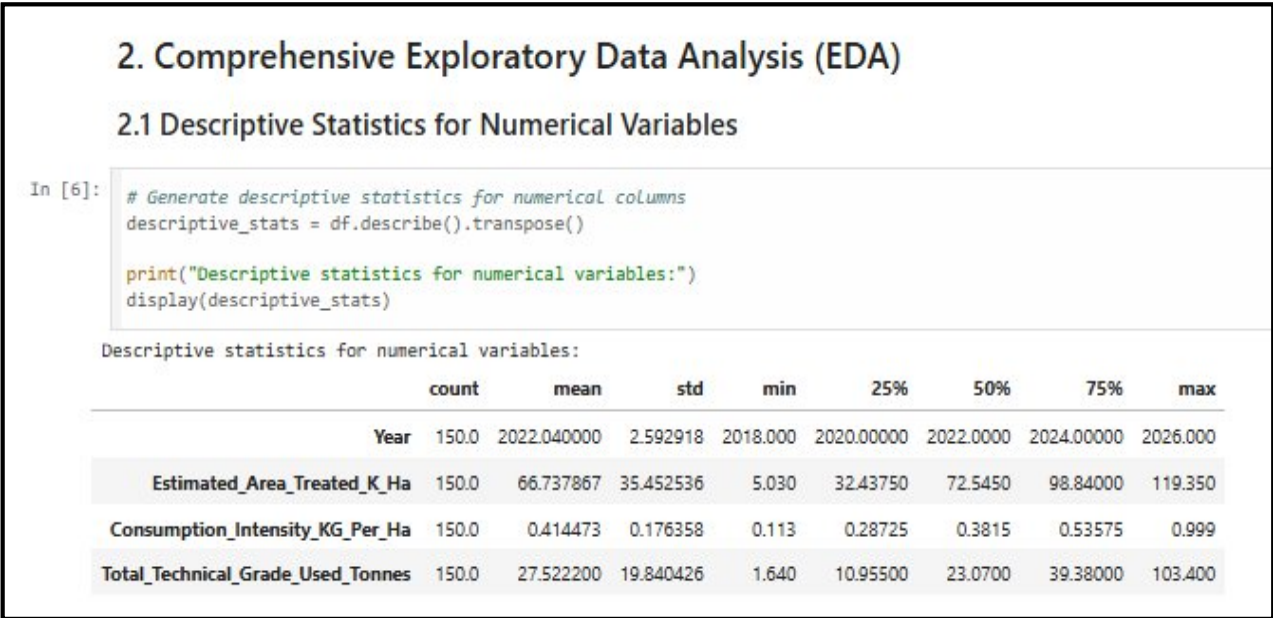
Screenshot 1: Notebook title and project introduction



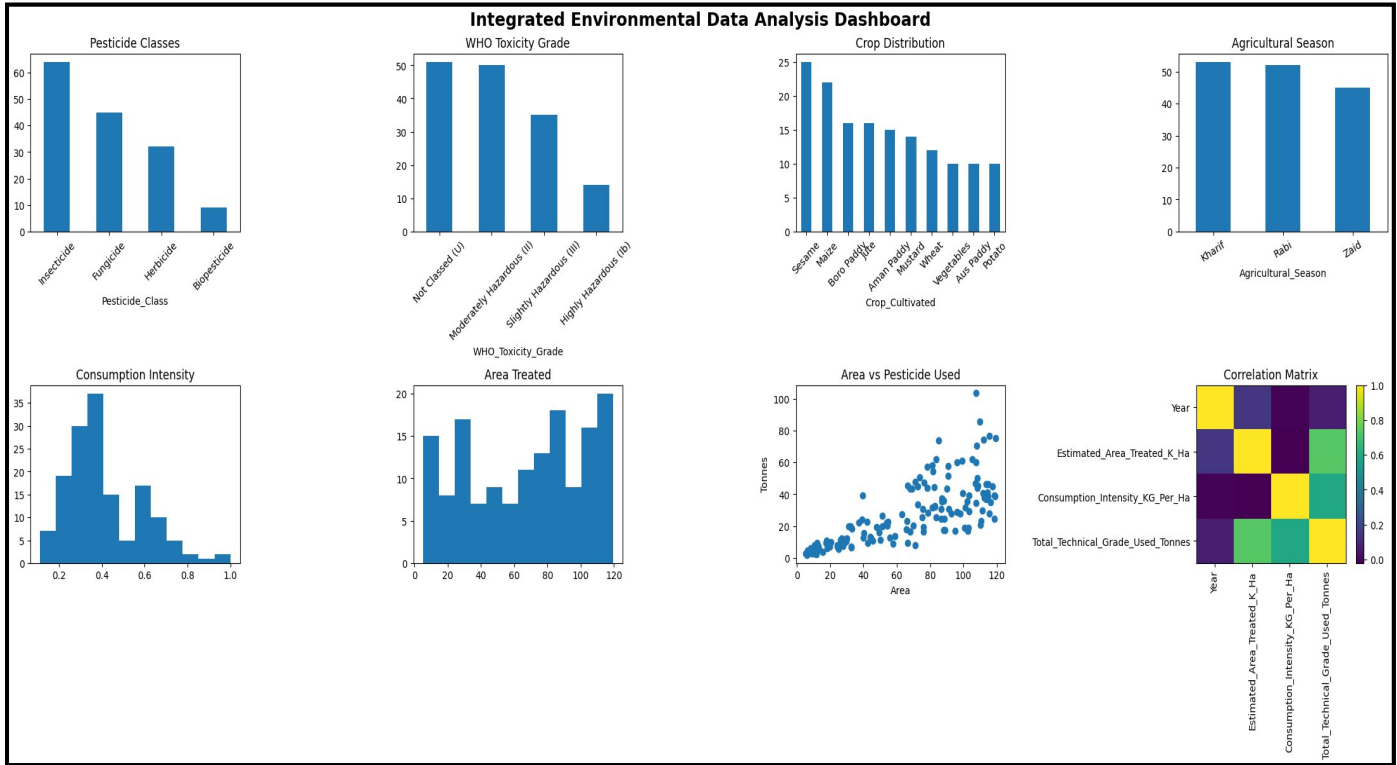
AI-Generated Results

The project successfully analyzed pesticide consumption data using Python and AI-assisted interpretation. Exploratory Data Analysis revealed variability in pesticide application intensity, treated agricultural area, and toxicity categories. The integrated dashboard summarized key trends, while ChatGPT generated structured environmental insights, recommendations for sustainable pesticide management, and potential applications of machine learning in environmental monitoring.

Screenshot 2 : Dataset loading and Exploratory Data Analysis



Screenshot 3 : Integrated Environmental Data Analysis Dashboard



Screenshot 4 : AI-Generated Environmental Assessment (Executive Summary or Key Findings)

Interpretation of the Integrated Environmental Dashboard

The integrated environmental dashboard provides a comprehensive overview of pesticide consumption patterns in West Bengal based on 150 observations collected between 2018 and 2026.

Key Findings

- **Pesticide Classes:** Insecticides are the most frequently represented pesticide class in the dataset, followed by fungicides and herbicides, while biopesticides constitute the smallest category. This suggests that insect pest management remains a major component of crop protection practices.
- **WHO Toxicity Profile:** The toxicity distribution indicates that **Not Classified (U)** (51 observations) and **Moderately Hazardous (Class II)** (50 observations) dominate the dataset, whereas **Highly Hazardous (Class Ib)** pesticides account for only 14 observations. Although highly hazardous pesticides are relatively uncommon, their continued presence highlights the need for careful regulation and safe handling.
- **Crop Distribution:** Multiple crops are represented in the dataset, with sesame and maize among the most frequently occurring. This demonstrates that pesticide usage is distributed across diverse cropping systems rather than concentrated on a single crop.
- **Agricultural Seasons:** Pesticide application records are relatively evenly distributed across the Kharif, Rabi, and Zaid seasons, indicating year-round pesticide usage associated with different cropping cycles.
- **Consumption Intensity:** Most observations are clustered around lower to moderate pesticide application rates, while a smaller number of records exhibit comparatively high consumption intensity. This indicates variability in pesticide management practices.
- **Agricultural Area Treated:** The treated agricultural area spans a wide range of values, reflecting differences in farm size, crop coverage, and cultivation practices among observations.
- **Relationship Between Area and Pesticide Usage:** The scatter plot reveals a clear positive relationship between the estimated agricultural area treated and the total technical-grade pesticide used. Larger treated areas generally require greater quantities of pesticides, although some variability suggests that application rates are also influenced by crop type and management practices.
- **Correlation Analysis:** The correlation matrix indicates a strong positive association between treated agricultural area and total technical-grade pesticide usage, while consumption intensity exhibits a moderate relationship with total pesticide use. The year variable shows comparatively weak correlations with the other numerical variables, suggesting that pesticide usage patterns are influenced more by agricultural characteristics than by temporal variation alone.

Overall Interpretation

The dashboard demonstrates considerable diversity in pesticide application across crops, seasons, toxicity classes, and agricultural areas. Together, these visualizations provide a strong exploratory foundation for AI-assisted environmental assessment and support further investigation using predictive analytics and machine learning techniques.

Learning Reflection

What I Learned

- Learned how Prompt Engineering improves the quality and relevance of AI-generated outputs.
- Developed practical experience using Google Collab for environmental data analysis.
- Applied Exploratory Data Analysis (EDA) using Python, Pandas, and Matplotlib.
- Created an integrated dashboard to visualize pesticide consumption patterns.
- Understood how AI can support scientific interpretation and environmental decision-making.

AI Tool I Liked Most

ChatGPT (GPT-5.5) was the most useful tool because it generated structured, context-aware, and scientifically meaningful analyses when provided with well-designed prompts.

How Prompt Engineering Improved the Output

Prompt Engineering enabled the AI to produce organized, domain-specific responses instead of generic summaries. By defining the AI's role, objectives, and expected report structure, the generated analysis became more accurate, coherent, and relevant to environmental science.
